PARMAR RAILWAY

COMPLETE GK PYQ BATCH

GENERAL KNOWLEDGE

(GEOGRAPHY)

LECTURE- 1

(BY- PARMAR SIR)

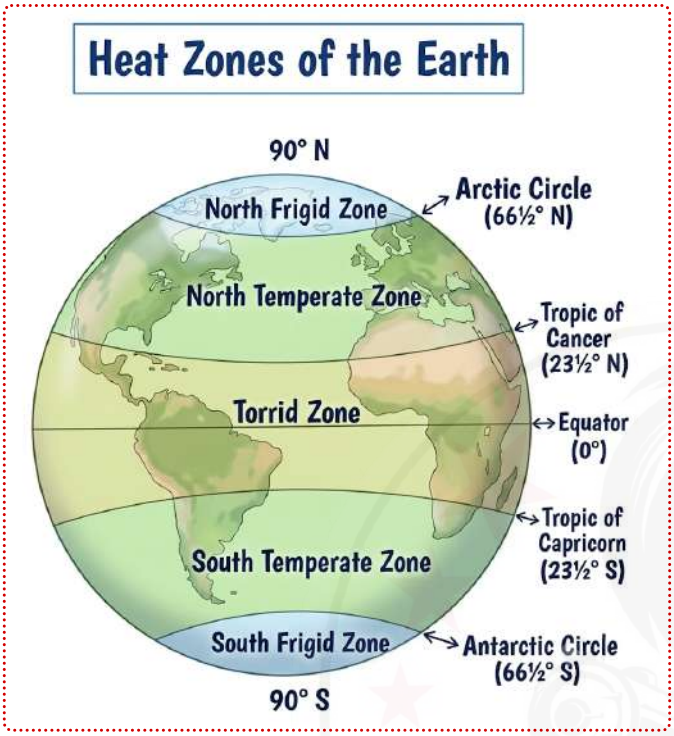


PHYSICAL GEOGRAPHY | INDIA GEOGRAPHY | WORLD GEOGRAPHY | ENVIRONMENT GEOGRAPHY | MAPPING CONCEPTS

Q1. The tropical climatic region is mainly confined to which latitudinal extent of the Earth?

[NTPC Graduation CBT-01 2025-26 (18 Mar, 2026 Shift 3)]

(A) 28.5°N and 28.5°S
(B) 23.5°N and 23.5°S
(C) 13.5°N and 13.5°S
(D) 30.5°N and 30.5°S



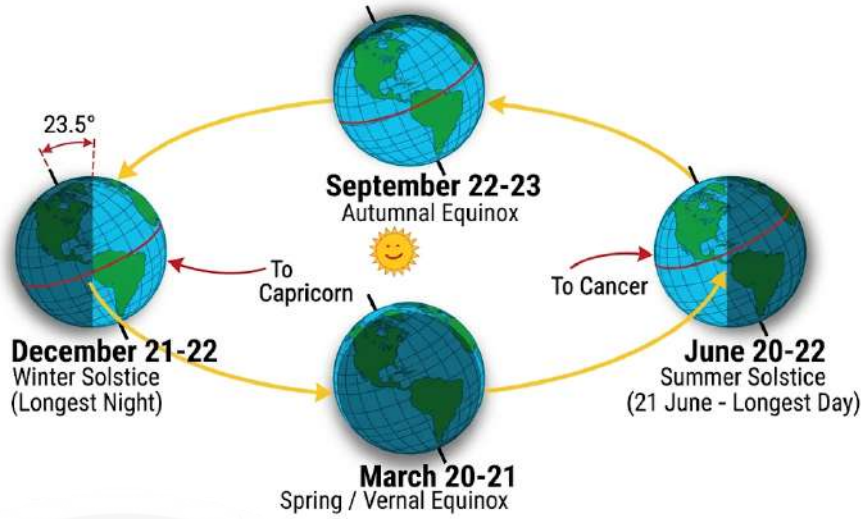
- **Tropical zone = Torrid Zone**
  - The area between the **Tropic of Cancer (23½° N)** and the **Tropic of Capricorn (23½° S)**
- The Sun's **direct (overhead) rays** fall only between these two latitudes
  - The Sun never goes north of 23½° N or south of 23½° S
  - Sunlight still reaches beyond these latitudes, but the Sun is never overhead there
- Direct rays give **maximum heat** in this belt, so it is called the **Torrid Zone**
- Beyond the tropics the rays are **slanting**
  - Slanting rays spread over a larger area, so the heat is less
  - Example: a convex lens brings rays to one point (small area) and the heat rises, so paper burns. Slanting rays are the opposite: large area, less heat

- **Latitude = Akshansh; Longitude = Deshantar**
- Both are **imaginary lines**, drawn to:
  - **Locate any place** on a map
  - Divide the Earth into different **climatic (heat)** zones
- Latitudes are horizontal lines; longitudes are **vertical** lines
  - They look like lines on a 2D map, but in reality they are **circles**
- **Latitudes**
  - Total number = **181** (90 north + 90 south + the Equator)
  - Equator = 0°; it divides the Earth into two equal halves: **Northern Hemisphere** and **Southern** Hemisphere
  - **90° N = North Pole; 90° S = South Pole**
  - Latitude circles get **smaller towards the poles**
  - The **Equator is the largest circle = Great Circle**
    - Great circle: the biggest circle, covering the full circumference of the Earth and dividing it into two equal parts
    - The shortest distance between two points lies along a great circle
  - Other latitudes (for example the Tropics) are not great circles
- **Longitudes**
  - Total number = 360
  - **Every longitude is a great circle**
- **Heat zones of the Earth**
  - **Torrid Zone**: 23½° N to 23½° S
  - **North Temperate Zone** and **South Temperate Zone**: between **23½°** and **66½°** in each hemisphere
    - Heat is less here (for example, European countries are not very hot)
  - **66½° N = Arctic Circle; 66½° S = Antarctic Circle**
  - **North Frigid Zone** and **South Frigid Zone**: from 66½° to the poles (90°)

Q2. Which latitude experiences the longest daylight during the June solstice?

[NTPC UG CBT-01 2025-26 (13 Jun, 2026, 4:30 PM - 6:00 PM)]

(A) 66½° N
(B) 23½° S
(C) 23½° N
(D) Equator



- Earth's axis is tilted **23½°** from the vertical; the axis makes **66½°** with the orbital plane (the plane of Earth's path around the Sun)
- Because of this tilt and Earth's **revolution** around the Sun:
  - At one point of the orbit the **Northern Hemisphere** leans towards the Sun
  - At another point the **Southern Hemisphere** leans towards the Sun
  - Twice a year neither hemisphere leans towards the Sun

- In the **June solstice (21 June)** the Northern Hemisphere is tilted towards the Sun
  - The region beyond **66½° N** stays in the sunlit part however much the Earth rotates
  - So there is **continuous daylight** (no darkness at night)
  - Hence the longest daylight in the north is at **66½° N**

- **Beyond 66½°**: about **6 months of day and 6 months of night**
  - When the Northern Hemisphere is tilted towards the Sun: continuous day beyond 66½° N, and darkness beyond 66½° S
  - In the **December solstice (22 Dec)** it is reversed: longest daylight at **66½° S**

- 23½° N (Option C) is the place of **overhead Sun** on 21 June, not the place of the longest daylight

- Shape of the Earth: **Geoid / Oblate Spheroid** (not a perfect sphere)

Q3. The summer solstice occurs when the Sun is directly overhead at which of the following latitudes?

NTPC Graduation CBT-01 2025-26 (16 Mar, 2026 Shift 1)]

(A) Arctic Circle
(B) Tropic of Capricorn
(C) Equator
(D) Tropic of Cancer

- **Summer solstice = 21 June**
  - The Northern Hemisphere is tilted towards the Sun
  - The Sun's direct (overhead) rays fall on the **Tropic of Cancer (23½° N)**
- Direct rays can **never** fall north of the Tropic of Cancer or south of the Tropic of Capricorn
  - So the **Arctic Circle** never gets overhead rays (Option A is wrong)
- **Winter solstice = 22 December**: direct rays on the **Tropic of Capricorn**

- **Equinox:** direct rays fall on the **Equator**
  - **21 March = Vernal (Spring) Equinox**
  - **23 September = Autumnal Equinox**
  - The entire Earth gets **equal day and night**
  - Memory tip: **Equinox = equal** day and night
- On **21 June: longest** day and **shortest night** in the Northern Hemisphere
- On **22 December: longest night** in the Northern Hemisphere
- Why 21 June is called "summer" solstice: most countries (including **India**) lie in the **Northern Hemisphere**, where it is summer at this time. This is a convention

Q4. Which of the following continents is NOT crossed by the Equator?

[NTPC Graduation CBT-01 2025-26 (16 Mar, 2026 Shift 3)]

(A) Africa
(B) South America
(C) Europe
(D) Asia

- The **Equator (0°)** passes through **Africa, South America and Asia**
  - In Asia it passes through **Maldives** and **Indonesia**
- It does **not** pass through **Europe**
- No need to memorise the full list of countries on the Equator; just know the continents



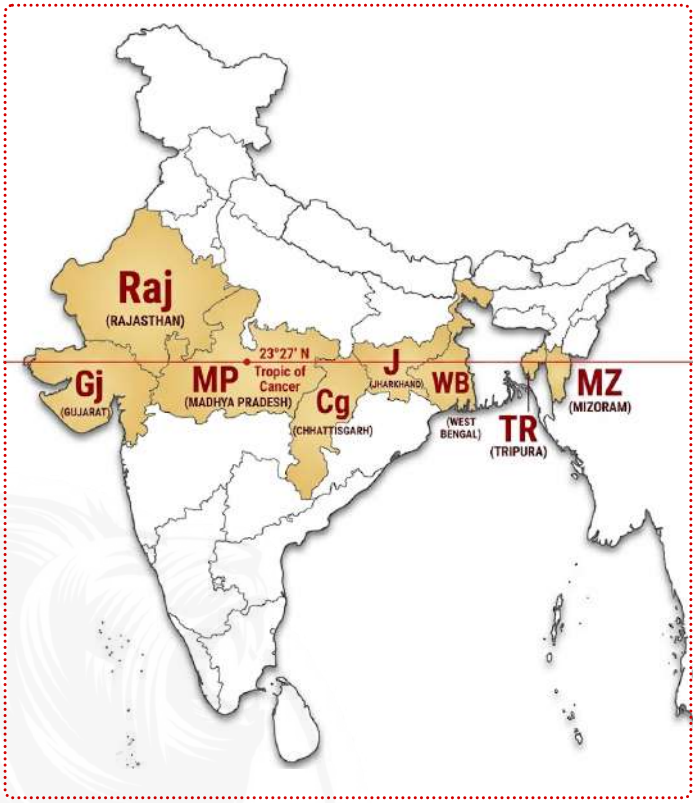
- **Tropic of Cancer (23½° N)** passes through **North America, Africa, Asia** (through the middle of India)
  - Does not pass through Australia or South America
- **Tropic of Capricorn (23½° S)** passes through **South America, Africa, Australia**
  - Does not pass through Asia
- **Equator and Tropic of Capricorn** both pass through **South America**
- **Africa** is the only continent crossed by **all three** (Equator, Tropic of Cancer, Tropic of Capricorn)
  - Africa is the "continent of continents": it has the **most countries**

Q5. Which of the following Indian states is NOT crossed by the Tropic of Cancer?

[NTPC UG CBT-01 2025-26 (13 Jun, 2026, 12:45 PM - 2:15 PM)]

(A) Odisha
(B) Gujarat
(C) Madhya Pradesh
(D) West Bengal

- The Tropic of Cancer (an imaginary line, **23½° N**) passes through the middle of India and crosses **8 states:**
  - **Gujarat → Rajasthan → Madhya Pradesh → Chhattisgarh → Jharkhand → West Bengal → Tripura → Mizoram**
- Odisha is not on the list
- Mnemonic: "**Gujarati Raja Made Chief Justice Win the Meeting**" (G-R-M-C-J-W-T-M)



- Map-based common sense helps: Bihar and Assam lie north of the Tropic of Cancer, so they are not crossed

Q6. The International Date Line approximately follows which meridian?

[RRB NTPC UG CBT-01 2024 (01 Sept, 2025 Shift 1)]

(A) 90°E
(B) 180°
(C) 75° W
(D) 0°

- The **International Date Line (IDL)** is the line where the **date changes**
- It follows the **180° meridian** (180° E and 180° W are the same line)
- It lies **opposite to the Prime Meridian**
- It passes through the **Pacific Ocean**
- It is **zigzag,** not straight
  - Reason: if it were straight, small island countries would be split and would have different dates on two sides
- Crossing rule
  - Crossing **westward** (from America side towards Japan / Asia side): **add one day** (you lose a day)
    - Example: Tuesday becomes Wednesday; 25 October becomes 26 October
  - Crossing **eastward**: the reverse (subtract one day)
- Few questions are asked on this, but it can be asked

- **Oceans (largest to smallest):** Pacific, Atlantic, Indian, Southern (Antarctic), Arctic
  - Memory tip: **PAISA** (Pacific, Atlantic, Indian, Southern, Arctic)
  - **Pacific** is the largest and also the **deepest** ocean
  - Deepest point: **Challenger Deep** in **Mariana Trench** (in the Pacific)
  - **Indian Ocean** is the only ocean named after a country
    - Deepest point in Indian Ocean - **Sunda Trench (Java Trench)**
  - **Arctic** is the smallest ocean
- **Atlantic Ocean** is the **busiest** ocean
  - It is S-shaped, with America, Europe, South America and Africa on its sides
- **Sargasso Sea** is in the North Atlantic
  - It gets its name from **Sargassum**, a brown algae (seaweed) that collects there
  - It is called a **borderless sea** (no land boundaries)

Q7. Which of the following represents the longitudinal extent of India?

[RRB NTPC UG CBT-01 2024 (12 Aug, 2025 Shift 3)]

(A) 68°7'E and 97°25'E
(B) 56°7'E and 87°25'E
(C) 58°7'E and 94°25'E
(D) 54°7'E and 77°25'E

- **Distance between two latitudes** is the same everywhere: **111 km**
- **Distance between two longitudes** is **not** the same
  - Maximum at the **Equator: 111.32 km**
  - Decreases towards the poles
  - **Zero at the poles**

- India's **longitudinal extent** (Deshantariya vistar): **68°7′ E to 97°25′ E**
- India's **latitudinal extent** (Akshansh vistar): **8°4′ N to 37°6′ N**
- India lies in the **Northern** and **Eastern** Hemispheres (north-eastern hemisphere)
  - The Equator passes south of India
  - The Equator divides the Earth north–south; the Prime Meridian divides it east–west

- Memory trick
  - Remember the lowest latitude **(8°)** and the westernmost longitude **(68°)**, then add about **30** to each
  - 68 + 30 = 98, take one less = **97°**
  - 8 + 30 = 38, take one less = **37°**
  - Minutes **(25′ and 6′)** have to be remembered separately

Q8. If it is 12 noon at IST (82°30′E), what time is it approximately at 97°30′E longitude?

[RRB NTPC Graduation CBT-01 2024 (18 Jun, 2025 Shift 1)]

(A) 1:20 PM
(B) 12:30 PM
(C) 11:00 AM
(D) 1:00 PM

- Time is calculated from a reference longitude
  - **East** of the reference: **add** time
  - **West** of the reference: **subtract** time
- Earth rotates **360°** in **24 hours**, so it is divided into **24 time zones**
  - **15° = 1 hour**
  - **1° = 4 minutes**
- Steps
  - Difference between 97°30′ E and 82°30′ E = **15° = 1 hour**
  - 97°30′ E is **east** of IST, so add 1 hour
  - 12:00 noon + 1 hour = **1:00 PM**

- **Prime Meridian = 0° longitude**, the reference for world time
  - Also called the **Greenwich Meridian**; time is called **GMT (Greenwich Mean Time)**
  - It passes through the **Greenwich Laboratory, London**
  - Chosen because **Britain** was a very powerful country at that time
- **IST** is based on **82°30′ E**
  - It is **5½ hours ahead of GMT**
  - Working: 82.5° ÷ 15 = 5½ hours (15° × 5 = 75° = 5 hours; remaining 7.5° = half an hour)
  - It is ahead because India lies **east** of Greenwich
- The **IST meridian passes through 5 states: Madhya Pradesh, Odisha, Uttar Pradesh, Chhattisgarh, Andhra Pradesh**
  - Mnemonic: **"MOUCA" (Mauka)**
  - Also called the **Mirzapur Clock Tower**, because it passes through **Mirzapur** and also through **Naini** (Allahabad / Prayagraj)

Q9. Among the following cities, which is closest to the Prime Meridian?

[RRB NTPC Graduation CBT-01 2024 (17 Jun, 2025 Shift 2)]

(A) Ahmedabad
(B) Agartala
(C) Kanyakumari
(D) Delhi

- The Prime Meridian lies to the **west** of India
- So the **westernmost** place among the options is the closest
  - **Ahmedabad** (Gujarat) is the westernmost
  - **Agartala** (Tripura) is in the far east
  - **Kanyakumari** is in the far south
  - **Delhi** is in the central part

- The Prime Meridian is a **straight line**
- It passes through the **Arctic Ocean, Atlantic Ocean and Southern Ocean** (not the Pacific)
- Because of this, questions on "which ocean the Prime Meridian passes through" are not asked

Q10. The 24° latitude passes closest to which Indian state's capital?

[RRB NTPC Graduation CBT-01 2024 (05 Jun, 2025 Shift 1)]

(A) Patna
(B) Bhopal
(C) Raipur
(D) Dispur

- The Tropic of Cancer **(23½° N)** passes through the middle of India
  - It crosses **Madhya Pradesh** (through the middle) and the northern part of **Chhattisgarh**
- **24° N** is only about half a degree north of it
  - So the capital closest to it must lie near the Tropic of Cancer
  - Bihar (Patna) and Assam (Dispur) are much farther north
  - **Raipur** (Chhattisgarh) is well south of the line, because the line runs along the upper part of the state
  - **Bhopal** (Madhya Pradesh) is the nearest

- The Tropic of Cancer cuts only a small part of **Rajasthan**
- The **24th parallel** passes through **Sir Creek** (India–Pakistan boundary area)
  - It was an option in a UP-level exam; it has not appeared in railway exams so far

Q11. Which neighbouring country is separated from India by the Palk Strait?

[NTPC UG CBT-01 2025-26 (20 Jun, 2026, 12:45 PM - 2:15 PM)]

(A) Sri Lanka
(B) Afghanistan
(C) Myanmar
(D) Maldives

- The **Palk Strait** lies between **India** and **Sri Lanka**
- India's boundary with neighbours
  - **Afghanistan** and **Myanmar**: **land** boundary
  - **Sri Lanka** and **Maldives**: **maritime (coastal)** boundary
- **Strait** = a narrow water body that separates (or connects) two larger water bodies

- Example: **Strait of Hormuz** connects the **Persian Gulf** and the **Gulf of Oman** (it was in the news)
- **Gulf of Mannar** also lies between India and Sri Lanka
- **Hambantota** Port (Sri Lanka) is on a **99-year lease** to **China**

Q12. The Garo, Khasi, Jaintia, and Naga Hills, which run almost east-west, are connected to which hill chain that extends in a north-south direction?

[NTPC Graduation CBT-01 2025-26 (22 Mar, 2026 Shift 1)]

(A) Mizo and Vindhya Hills
(B) Mizo and Rakhine Hills
(C) Satpura and Mizo Hills
(D) Nilgiri and Rakhine Hills

- **Garo, Khasi, Jaintia** (Meghalaya) and **Naga Hills** run **east-west**
- **Mizo Hills** and **Rakhine Hills** (also called **Arakan Yoma**, in Myanmar) run **north-south**

- Eliminating the wrong options
  - **Vindhya** and **Satpura** run east-west, with the **Narmada** flowing between them
  - **Nilgiri** is in the south (border of Kerala, Karnataka and Tamil Nadu), far from Meghalaya

Q13. Which of the following countries serves as India's gateway to the ASEAN region by being the only ASEAN nation sharing a land boundary with India?

[NTPC Graduation CBT-01 2025-26 (18 Mar, 2026 Shift 1)]

(A) Myanmar
(B) Bangladesh
(C) Nepal
(D) Bhutan

- Old name of Myanmar: **Burma**
- Capital of Myanmar: **Naypyidaw**

- **Myanmar** is the **only ASEAN member** that shares a land boundary with India
- **Bangladesh, Nepal and Bhutan** are **not** ASEAN members
- **ASEAN** = Association of South East Asian Nations
  - Established in **1967** at **Bangkok, Thailand**
  - Headquarters: **Jakarta, Indonesia**
- Indian states bordering Myanmar: **Arunachal Pradesh, Nagaland, Manipur, Mizoram**
  - Mnemonic: "**Aru-Na-Ma-Mi**"

Q14. Which island country, along with Sri Lanka, is identified as India's southern neighbour across the Indian Ocean?

[NTPC Graduation CBT-01 2025-26 (16 Mar, 2026 Shift 3)]

(A) Thailand
(B) Indonesia
(C) Maldives
(D) Bhutan

- **Maldives**
  - Capital: **Male**
  - Language: **Dhivehi**
  - President: **Mohamed Muizzu**
- **Lakshadweep**
  - **Smallest Union Territory** of India
  - Total area: about **32.6 sq km**
  - **Smallest island: Bitra** (area about **0.105 sq km**)
  - **Largest island: Andrott** (area about **4.98 sq km**)
  - **Southernmost island: Minicoy**

- Only **two** neighbours have a **coastal (maritime) boundary** with India: **Sri Lanka** and **Maldives**
- India has two island groups
  - **Lakshadweep** (Arabian Sea)
  - **Andaman and Nicobar** (Bay of Bengal)
- **Minicoy** is the southernmost island of Lakshadweep; after it comes **Maldives**
- The **8° Channel** (8° N latitude) separates Minicoy / Lakshadweep from Maldives

Q15. Which of the following States/Union Territories shares a border with Afghanistan?

[NTPC Graduation CBT-01 2025-26 (17 Mar, 2026 Shift 1)]

(A) Rajasthan
(B) Ladakh
(C) Jammu and Kashmir
(D) Punjab

- Memory tip: both start with "L"
  - Lakshadweep = smallest UT
  - Ladakh = largest UT

- **Ladakh** (Union Territory) shares a border with **Afghanistan**
- Ladakh is the **largest Union Territory** of India

Q16. Which Union Territory of India is located in the Arabian Sea?

[RRB NTPC UG CBT-01 2024 (03 Sept, 2025 Shift 3)]

(A) Andaman and Nicobar Islands
(B) Chandigarh
(C) Lakshadweep
(D) Puducherry

- **Lakshadweep** is in the **Arabian Sea** (Arab Sagar)
- **Andaman** and **Nicobar Islands** are in the **Bay of Bengal**

- **Puducherry** has the **highest sex ratio among Union Territories: 1037**

Q17. The McMahon Line is the boundary between India and which country?

[RRB NTPC UG CBT-01 2024 (02 Sept, 2025 Shift 3)]

(A) Pakistan
(B) Myanmar
(C) Afghanistan
(D) China

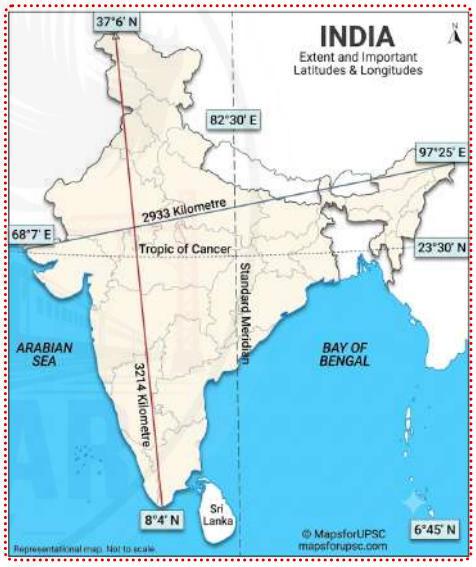
- **McMahon Line** = boundary between **India and China**

- **Radcliffe Line** = boundary between **India and Pakistan**, and also **India and Bangladesh**
  - Drawn by **Sir Cyril Radcliffe**; Bangladesh did not exist then (it was **East Pakistan**)
- **Durand Line** = boundary between **Pakistan and Afghanistan**

Q18. What is the actual mainland distance of India from east to west?

[RRB NTPC UG CBT-01 2024 (02 Sept, 2025 Shift 3)]

(A) 2933 km
(B) 3143 km
(C) 3214 km
(D) 3014 km



- **East-west distance** (Gujarat to Arunachal Pradesh, along the mainland): **2933 km**
- **North-south distance** (Ladakh to Kanyakumari): **3214 km**
- North-south extent is **greater** than east-west extent

Q19. Which islands of India are strategically important due to their location near the Malacca Strait?

[RRB NTPC UG CBT-01 2024 (21 Aug, 2025 Shift 2)]

(A) Gulf of Kutch Islands
(B) Andaman and Nicobar Islands
(C) Lakshadweep Islands
(D) Rameswaram Islands

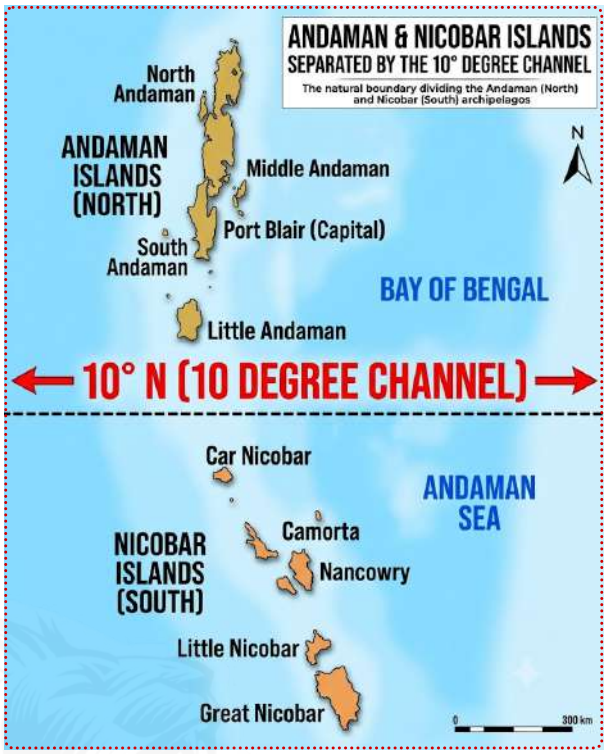
- The **Malacca Strait** connects the **Andaman Sea** and the **South China Sea**
- It is very important for trade
- The **Andaman and Nicobar Islands** lie near it, so they are strategically important

Q20. The Andaman and Nicobar Islands are separated by _____.

[RRB NTPC UG CBT-01 2024 (20 Aug, 2025 Shift 3)]

(A) 12° N parallel
(B) 10° N parallel
(C) 14° N parallel
(D) 16° N parallel

- The **10° N latitude** passes between the two groups, so it is called the **10° Channel**
  - Everything **above** it is **Andaman**
  - Everything **below** it is **Nicobar**
- It separates **Andaman from Nicobar**, and specifically **Little Andaman from Car Nicobar**
- Islands
  - **Andaman**: North Andaman, Middle Andaman, South Andaman, Little Andaman
  - **Nicobar**: Car Nicobar, Little Nicobar, Great Nicobar
- **Duncan Passage** separates **South Andaman** from **Little Andaman**



- **Highest peak: Saddle Peak** (North Andaman), height 732 m
  - 2nd highest: **Mount Thuillier**
  - 3rd highest: **Mount Diavolo**
  - 4th highest: **Mount Koyob**
  - Memory tip: 1st at the top, 2nd at the bottom, then 3rd and 4th going back up

- **Only active volcano of India: Barren Island**
- **Only dormant volcano: Narcondam** (slightly east of North Andaman)
- **Capital: Sri Vijaya Puram** (earlier Port Blair), located in South Andaman
- **Total area: 8249 sq km**

Q21. Which Indian state shares its border with China, Nepal and Bhutan?

[RRB NTPC UG CBT-01 2024 (12 Aug, 2025 Shift 3)]

(A) Himachal Pradesh
(B) Uttarakhand
(C) Arunachal Pradesh
(D) Sikkim

- Only **three Indian states** share borders with **three countries** each
  - **West Bengal**: Nepal, Bhutan, Bangladesh
  - **Arunachal Pradesh**: Bhutan, China (Tibet), Myanmar
  - **Sikkim**: China (Tibet), Bhutan, Nepal
- China + Nepal + Bhutan = **Sikkim**

- **Ladakh** (a UT, not a state) also borders three countries: **Afghanistan, Pakistan, China**
  - Places in the Ladakh region shown on the map: **Galwan Valley**
  - **Aksai Chin** is under Chinese control (India claims it)
  - **PoK** is under illegal Pakistani occupation
- Sikkim (polity link)
  - **35th Amendment, 1974**: Sikkim made an Associate State; Article 2A inserted
  - **36th Amendment, 1975**: Sikkim got **full statehood**
  - **Article 371F**: special provisions for Sikkim
  - Earlier ruled by the **Chogyal dynasty**
  - **Least populated state** of India (by population, not by area)

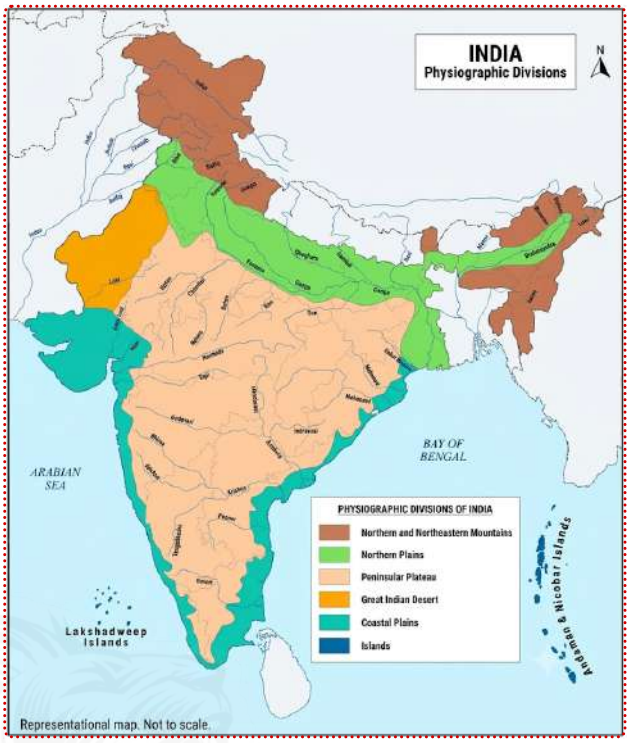
Q22. What is the approximate area covered by the Thar Desert in India?

[RRB NTPC Graduation CBT-01 2024 (24 Jun, 2025 Shift 1)]

(A) 1,50,000 km²
(B) 2,00,000 km²
(C) 1,70,000 km²
(D) 1,00,000 km²

- Area of the Thar Desert in India: about **1,70,000 sq km**
- The **most densely populated desert in the world** (highest population density, i.e. most people per sq km)

- India is divided into **6 physiographic divisions**
  - The **Himalayas** (northern mountains)
  - The **Northern Plains** (alluvial / flood plains)
  - The **Peninsular Plateau**
  - The **Coastal Plains**
  - The **Group of Islands**: Lakshadweep and Andaman & Nicobar
  - The **Thar Desert**
  - The order is not important; the list is not numbered by rank



Q23. The Kalapani dispute is between India and which of these?

[RRB NTPC Graduation CBT-01 2024 (23 Jun, 2025 Shift 2)]

(A) China
(B) Bangladesh
(C) Bhutan
(D) Nepal

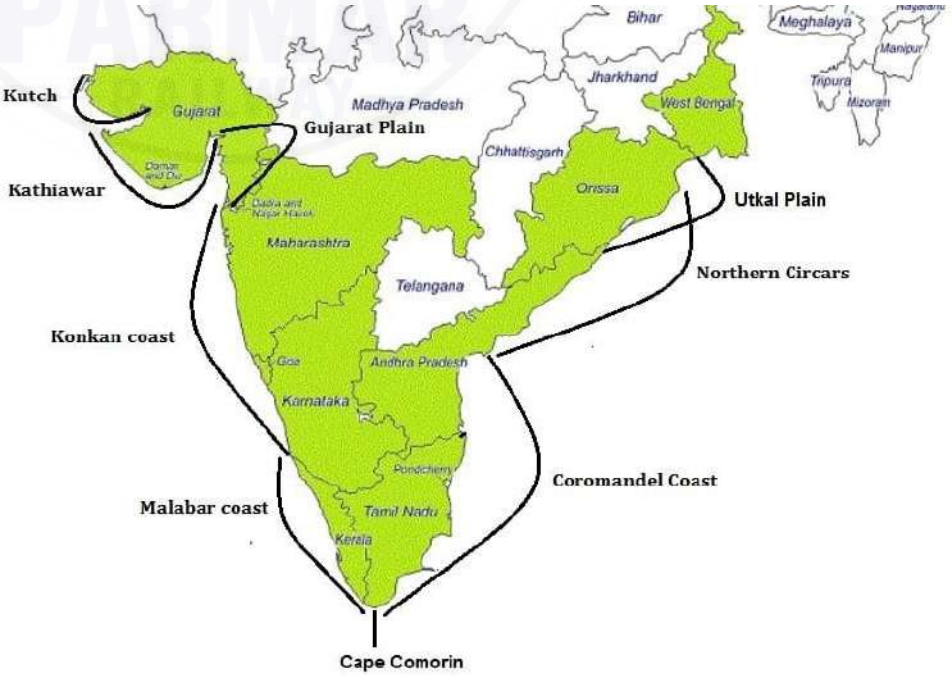
- **Kalapani, Lipulekh** and **Limpiyadhura** are disputed territories between **India and Nepal**

Q24. The Coromandel Coast is located along which part of India?

[RRB NTPC Graduation CBT-01 2024 (21 Jun, 2025 Shift 2)]

(A) Southern West coast
(B) Southern East coast
(C) Northern East coast
(D) Northern West coast

- **Coromandel Coast** = **Tamil Nadu** coast, on the **south-eastern** side of India
- **Malabar Coast** = **Kerala coast** (south-western side)
- These two (Coromandel and Malabar) are the **most important** names



- Coastal plains of the Peninsular Plateau
  - The **Western Ghats** are the western-most extension of the Peninsular Plateau
  - The **Eastern Ghats** are the eastern-most extension
  - **Western Coastal Plain:** between the **Western Ghats and the Arabian Sea**
  - **Eastern Coastal Plain:** between the **Eastern Ghats and the Bay of Bengal**
- **Names of coastal plains**
  - **Kutch Coast** and **Kathiawar Coast:** Gujarat
  - **Konkan Coast:** Maharashtra
  - **Kannad Coast:** Karnataka
  - **Malabar Coast:** Kerala
  - **Coromandel Coast:** Tamil Nadu
  - **Northern Circars:** Andhra Pradesh
  - **Utkal Coast:** Odisha
- **Western vs Eastern Coastal Plain** (full differences will be taught later)
  - Western: **narrow, submerging,** called a **retreating coast**
  - Eastern: **wide, emerging**
  - Most ports on the wide eastern coast are **artificial** in nature

Q25. The Lipulekh Pass, important for India's border trade, lies between India and _____.

[RRB NTPC Graduation CBT-01 2024 (18 Jun, 2025 Shift 2)]

(A) Bangladesh
(B) Bhutan
(C) Tibet (China)
(D) Afghanistan



- **Passes in Uttarakhand: Niti, Mana, Muling La, Mangsha Dhura, Lipulekh**
  - Memory aid used by the teacher: "Mooli maang ke laaya"
- **Passes in Sikkim:** only two, **Nathu La** and **Jelep La**
- **Passes in Arunachal Pradesh: Bum La, Bomdi La, Yonggyap, Dihang, Diphu**
- Detailed study of passes will come with the Himalayas lecture

- **Lipulekh Pass** is in **Uttarakhand**
- It is a **tri-junction** of three countries: **India, Nepal and Tibet**
- Nepal is not given in the options, so the answer is **Tibet (China)**

Q26. What is the name of the point where India, China and Myanmar meet?

[RRB NTPC Graduation CBT-01 2024 (16 Jun, 2025 Shift 2)]

(A) Diphu Pass
(B) Anjaw Point
(C) Kibithu
(D) Tri-Junction Pass

- **Diphu Pass** (Arunachal Pradesh) is the **tri-junction** where **India, China** and **Myanmar** meet
- Similar tri-junction idea: **Lipulekh** = India, Nepal, Tibet

Q27. Most of the Thar desert lies within India, spanning the states of _____

[RRB NTPC Graduation CBT-01 2024 (13 Jun, 2025 Shift 1)]

(A) Rajasthan, Gujarat, Punjab, and Uttar Pradesh
(B) Rajasthan, Gujarat, Punjab, and Haryana
(C) Rajasthan, Gujarat, Punjab, and Madhya Pradesh
(D) Rajasthan, Gujarat, Punjab, and Himachal Pradesh

- **Ladakh** is also a desert, but **a cold desert**

- The Thar Desert is in the **western / north-western** part of India, **west of the Aravalli range**
- It spreads over **Rajasthan, Gujarat, Punjab and Haryana**
- Eliminating options
  - **Himachal Pradesh:** too cold; Thar is a **hot desert**
  - **Madhya Pradesh** and **Uttar Pradesh:** not in the desert region

Q28. Which of these is the westernmost inhabited village in India?

[RRB NTPC Graduation CBT-01 2024 (12 Jun, 2025 Shift 2)]

(A) Lakhpat
(B) Tanot
(C) Koteshwar
(D) Ghuar Mota

- **Ghuar Mota (Guhar Moti) in Gujarat** = westernmost inhabited village of India

- **Extreme points of India**
  - **Easternmost village: Kibithu** (Arunachal Pradesh)
  - **Northernmost point: Indira** Col (Ladakh)
  - **Southernmost point of the mainland: Cape Comorin (Kanyakumari)**
  - **Southernmost point overall: Indira Point** (Great Nicobar, Andaman & Nicobar)
    - It was submerged in the **2004 Boxing Day tsunami**

- **Time difference within India**
  - Longitude difference between Gujarat and Arunachal: about **30°** (68° to 97°)
  - 15° = 1 hour, so about **2 hours;** Arunachal is ahead
  - Clocks show the same time everywhere in Indiabe cause **IST (82½° E)** is followed for the whole country
  - East is ahead because the **Sun rises from the east**
- **Japan** is known as the **Land of the Rising Sun**
- **Norway** is known as the **Land of Midnight Sun**

Q29. India's territorial waters extend up to how many nautical miles from the baseline?

[RRB NTPC Graduation CBT-01 2024 (11 Jun, 2025 Shift 3)]

(A) 200
(B) 100
(C) 12
(D) 24

- **UNCLOS** (United Nations Convention on the Law of the Sea) divides marine areas into 5 zones, measured from the baseline (coast line):
  - **Internal Waters:** waters inside the land (rivers, lakes, ponds); full control of the country
  - **Territorial Sea:** up to 12 nautical miles (about 22 km)
  - **Contiguous Zone:** up to 24 nautical miles
  - **Exclusive Economic Zone (EEZ):** up to 200 nautical miles
  - **High Seas:** everything beyond the EEZ
- **1 nautical mile ≈ 1.85 km**

- **EEZ:** the country can do **research** and **exploit resources** (for example, polymetallic nodules)
- **High Seas** are a **common heritage**; India cannot claim them
- **Coastline of India**
  - Earlier figure: **7516 km** (7516.6 km)
  - Updated total (NHO and Survey of India, 2023-24): **11,098.81 km**
    - Mainland: **7,870.51 km**
    - Islands: **3,228.30 km**
  - **9 states** and 4 **Union Territories** have a coastline
  - Longest coastline: Union Territory **Andaman and Nicobar (3083 km)**
  - **Longest among states: Gujarat (2340 km)**
    - 2nd: **Tamil Nadu**
    - 3rd: **Andhra Pradesh**
    - Memory tip: **GTA** (Gujarat, Tamil Nadu, Andhra Pradesh)
  - **Shortest coastline:** state **Goa**; UT **Puducherry**

Q30. Which of the following is not a surrounding feature of the Thar Desert?

[RRB NTPC Graduation CBT-01 2024 (06 Jun, 2025 Shift 2)]

(A) Rann of Kachchh
(B) Aravalli Range
(C) Vindhya Hills
(D) Indus River Plain

- **Vindhya Hills** are far away from the Thar Desert

- Features around the Thar Desert
  - **Rann of Kachchh** (towards the south)
  - **Aravalli Range**
  - **Indus River** Plain (the Indus and its tributaries reach up to this region)

Q31. The India-Bhutan boundary is primarily defined by which of these?

[RRB NTPC Graduation CBT-01 2024 (06 Jun, 2025 Shift 1)]

(A) Desert
(B) Mountains and treaties
(C) Trade posts
(D) Border fencing

- The India–Bhutan boundary is defined by **mountains and treaties**
- **Nepal** and **Bhutan** are **peaceful, open (porous) borders**
  - Guarded by **Sashastra Seema Bal (SSB)**
  - Not like the Pakistan and China borders, so there is no border fencing here

- **India's land boundary** is shared with **7 countries**
  - Mnemonic: "**Bachpan mein MBA**" = Bangladesh, China, Pakistan, Nepal, Myanmar, Bhutan, Afghanistan
  - This is the **decreasing order** of border length
  - **Total international land boundary: 15,106 km**
  - **Bangladesh:** longest, **4096.7 km**
  - **China: 3488 km**
  - **Pakistan: 3323 km** (memory tip: repeated 3s)
  - **Nepal: 1751 km**
  - **Myanmar: 1643 km**
  - **Bhutan: 699** km
  - **Afghanistan:** shortest, **106 km**
  - No need to memorise all the lengths; the order is more important

- **States bordering Bhutan: West Bengal, Assam, Sikkim, Arunachal Pradesh**
  - Mnemonic: "**BASA**"

- **States bordering Nepal: Sikkim, Uttar Pradesh, Bihar, West Bengal, Uttarakhand**
  - Mnemonic: "**SUBBU**" (S-U-B-B-U)

- **States bordering Bangladesh: Meghalaya, Assam, Mizoram, Tripura, West Bengal**
  - Mnemonic: "**Mam Mam take water**" (M-A-M-T-W)
  - **West Bengal** has the **longest border** with Bangladesh among states

- **Tripura** is surrounded by Bangladesh on **three sides**
- **Chicken Neck Corridor**
  - The narrow part of **West Bengal** between Nepal, Bhutan and Bangladesh
  - Strategically very sensitive: if it is cut, the North-East gets disconnected from the rest of India
  - **Kaladan Multimodal Project** (Myanmar) links **Sittwe Port** with India, giving **indirect access to the North-East**

- Bhutan
  - Known as the **Land of the Thunder Dragon**
  - High on the **Happiness Index**
  - Famous for **hydroelectric power**

Q32. Which statement accurately explains the nature of the Inter Tropical Convergence Zone (ITCZ)?

[NTPC UG CBT-01 2025-26 (16 Jun, 2026, 12:45 PM - 2:15 PM)]

(A) It marks a westerly wind system close to the Tropic of Capricorn.
(B) It forms a cold frontal region near the Himalayan ranges.
(C) It is an equatorial low-pressure belt where trade winds meet.
(D) It represents a zone of high pressure located near the poles.

- **ITCZ = Inter Tropical Convergence Zone**
  - The zone where the **trade winds meet (converge)**
  - Trade winds: **North-Easterly** (Northern Hemisphere) and **South-Easterly** (Southern Hemisphere)
    - A wind is named after the direction from **which it blows**
  - It is an **equatorial low-pressure belt**
  - The same region is called the **Doldrums**
- Reason for low pressure: the Equator gets very high heat, so air **rises**

- **Wind basics**
  - Wind always blows from **high pressure to low pressure**
  - **Low pressure** = warm area, air rises
  - **High pressure** = cold area, air sinks (subsides)
- **Types of winds: Permanent (planetary) winds** and **Local winds**
  - Planetary winds cover the whole planet; local winds blow only in local areas
- **3 types of planetary winds: Trade Winds, Westerlies, Polar Easterlies**
- **Circulation cells** (in each hemisphere): **Hadley cell, Ferrel cell, Polar cell**
  - Air rises at the Equator, cools, and sinks at about 30° (Hadley cell)
- **Pressure belts**
  - **Equatorial Low** (ITCZ / Doldrums)
  - **Sub-tropical High** at **30° N** and **30° S** **(Horse Latitudes)**: air sinks here
  - **Sub-polar Low** at about **60°**
  - **Polar High** at the poles

- **Westerlies of the Southern Hemisphere:** very strong because there is **less landmass** in the southern hemisphere
  - **Roaring Forties** (40° latitude)
  - **Furious Fifties** (50° latitude)
  - **Shrieking Sixties** (60° latitude)
- **Coriolis Force** (also called **Ferrel's Law**)
  - Caused by the **rotation of the Earth**
  - Deflects winds to the **right in the Northern Hemisphere** and to the **left in the Southern Hemisphere**
  - This is why winds do not blow straight from high to low pressure; they curve (for example, trade winds become North-Easterlies and South-Easterlies)

Q33. Which of the following is a depositional landform formed by wind action?

[NTPC UG CBT-01 2025-26 (16 Jun, 2026, 4:30 PM - 6:00 PM)]

(A) Yardang
(B) Barchan
(C) Inselberg
(D) Moraine

- **Barchan** is a **sand dune** formed by wind **deposition**
- **Moraine** is formed by **glaciers**, not wind
- **Yardang** and **Inselberg** are **erosional** wind landforms

- Landforms are made by **wind, water, glaciers and sea water**
  - Two types: **Erosional** (made by erosion) and Depositional (made by deposition)
  - Landforms made by wind are called **Aeolian landforms** (not very important)
- **Why wind erodes in deserts**: desert sand is **coarse-grained**, and wind keeps hitting rocks with sand again and again, so the rocks wear away
- **Wind erosional landforms: Yardang, Inselberg, Monadnock, Mushroom Rock, Pediplain, Deflation Hollow**
  - **Pediplain** is formed by wind; **Peneplain** is formed by water
  - Mushroom rock has a mushroom shape

- **Wind depositional landforms** (all are types of **Sand Dunes**)
  - **Barchan:** has two wings pointing in the wind direction
  - **Seif:** only one wing
  - **Parabolic dune:** forms where there is vegetation cover; wings point opposite to the wind direction
  - **Transverse dune**

Q34. Which wind system blows consistently from high-pressure to low-pressure areas year-round?

[NTPC Graduation CBT-01 2025-26 (22 Mar, 2026 Shift 1)]

(A) Local winds
(B) Planetary winds
(C) Polar winds
(D) Periodic winds

- **Planetary (permanent) winds** blow **all year round** from high pressure to low pressure
  - Examples: Trade Winds, Westerlies, Polar Easterlies
- **Periodic** winds blow only at times (not all year)
- **Local** winds blow only in local areas, sometimes

- **Local winds**
  - **Loo:** a hot summer wind; blows in summer, not all year; a result of **advection (horizontal transfer of heat)**
  - **Chinook:** blows in **North America (USA)**; called "**Snow Eater**" (it melts snow), so it is a **warm** wind
  - **Bora and Mistral:** cold winds of **Europe**
  - **Sirocco:** blows in **Africa;** it lifts sand from the **Sahara**, causing **blood rain** (red-coloured rain)
  - **Harmattan:** blows in **Africa** on the **Guinea Coast**; known as the **Doctor Wind**
  - **Khamsin:** blows in **Egypt**

Q35. The cold and chilly winds that are blocked by the Himalayas. originate from Central Asia and regions close to which of the following?

[NTPC Graduation CBT-01 2025-26 (24 Mar, 2026 Shift 1)]

(A) Central Africa
(B) Pacific Ocean
(C) Arctic Circle
(D) Mediterranean Region

- Very cold winds come from the **Arctic Circle (66½° N)** side
- The **Himalayas block** them, so **North India** does not become extremely cold
  - Without the Himalayas the cold would be much more severe

- This will be discussed again in the Climate of India lecture

Q36. What is the low pressure belt near the equator commonly called?

[RRB NTPC UG CBT-01 2024 (20 Aug, 2025 Shift 3)]

(A) Polar Front
(B) Horse Latitudes
(C) Doldrums
(D) Trade Wind Belt

- **Horse Latitudes (Ashwa Akshansh):** winds **subside** (sink) here

- **Doldrums** = the **low-pressure belt near the Equator**, where the air is calm
- Sailors and traders gave this name because they found **almost no wind** there
- Eliminating options
  - **Horse Latitudes** = **30° N and 30° S** (high pressure)
  - **Trade Wind Belt** = around the Equator, but it is a much **larger belt**
  - **Polar Front** = not near the Equator

Q37. What is the typical direction of jet streams influencing the Indian climate in winter?

[RRB NTPC Graduation CBT-01 2024 (18 Jun, 2025 Shift 3)]

(A) East to west
(B) Southwest to northeast
(C) Southeast to northwest
(D) West to east

- Jet streams that affect India's winter climate blow from **west to east**
- Detailed explanation will come in the Climate of India (monsoon) lecture

Q38. What is the significance of Westerly Jet Streams in winter?

[RRB NTPC Graduation CBT-01 2024 (06 Jun, 2025 Shift 3)]

(A) Cause retreat of monsoon
(B) Influence rainfall in Tamil Nadu
(C) Guide Western Disturbances into India
(D) Bring snowfall to Eastern Ghats

- **Westerly jet streams bring the winter rain** of North India from the west
- Detailed study will come later in the Climate lecture\n



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